Scalable Transformer-Based Multi-Objective Contrastive Deep Reinforcement Learning for Molecular Optimization

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REFERENCES

Introduction

- Designing novel pharmaceutical-grade drugs is an extremely lengthy, costly, and intricate multi-objective optimization problem, as candidate molecules must be safe and efficacious, while satisfying several target properties [1],[2].
- The chemical search space for potentially synthesizable, drug-like molecules is estimated to be of size 10^{20}-10^{200} possible molecules [3].
  - **Deep reinforcement learning (DRL)** methods have achieved promising results in terms of efficiently navigating the incredibly vast search space [4]-[7]. However, factors such as lengthy training times and inefficient use of training data hinder their scalability and the quality of generated molecules [8].
- In this study, we propose a novel DRL framework for drug design based on Bidirectional Encoder Representations from Transformers (BERT) and contrastive learning. We show that the model has the ability to generalize its optimization capabilities to molecules not present in the pre-training or fine-tuning phases.

Methods

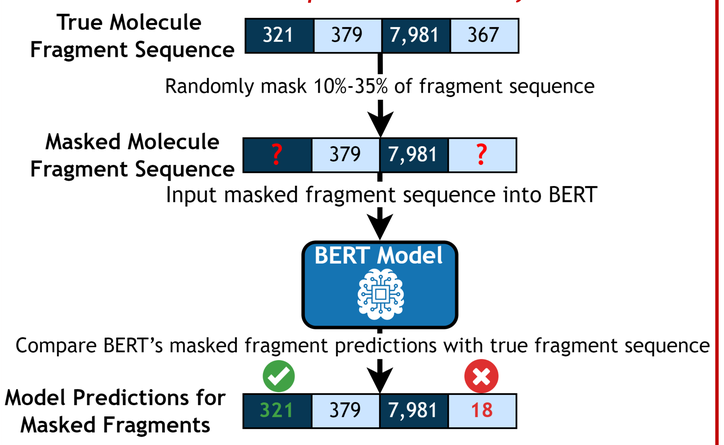
1. Pre-processing

Molecules (in the form of SMILES strings) of a 4 million molecule database [9] are fragmented via the HierVAE method [10] to generate a fragment dictionary, which enables the representation of molecules as numerical sequences.

Fragment Dictionary	
0	O=C1C
1	CCC
2	([Yb])=C
...	...
41,386	([Ta])C2SC

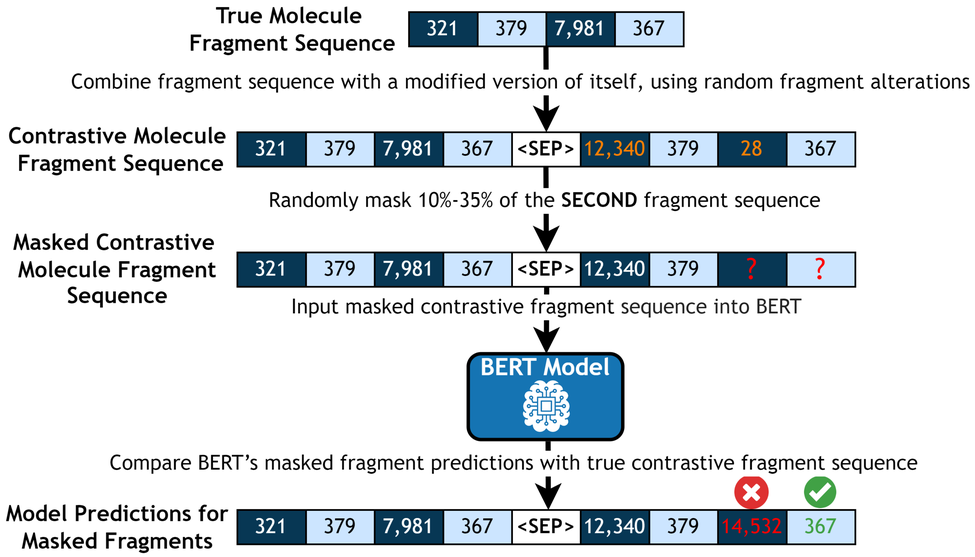
2.1. Pre-training: Masked Language Model (MLM)

Learn contextual representations of molecules



2.2. Pre-training: Contrastive Learning (CL)

Refine contextual representations by learning molecular similarities



3. Fine-tuning: Deep Reinforcement Learning (DRL)

Apply learned contextual representations for molecular optimization

Markov Decision Process: $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$:

$\mathcal{S}$: State space $\quad \mathcal{A}$: Action space $\quad \mathcal{P}$: State transition probability $\mathcal{P}(s_{t+1} \mid s_t, a_t)$

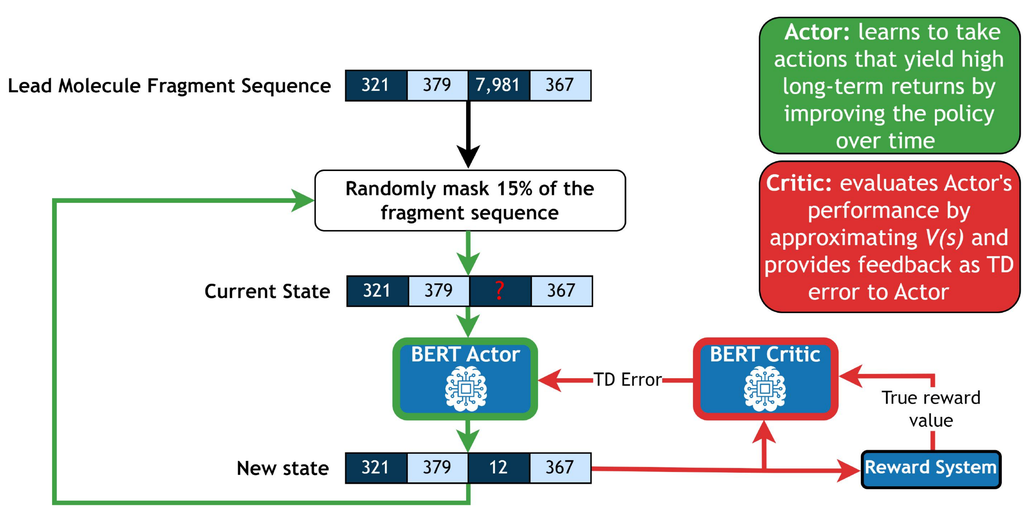
$\mathcal{R}$: Immediate reward probability $\mathcal{R}(r_t \mid s_t, a_t)$ $\quad \gamma \in [0,1)$: Discount factor

Policy: A distribution $\pi(a_t \mid s_t)$ which maps states to actions

State-Value Function: Evaluates the value of a state s under policy π :

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \sum_{s',r} p(s', r \mid s, a)\left[r + \gamma V^{\pi}(s')\right]$$

Temporal Difference (TD) Error: $(V(s_t) - (r_t + \gamma V(s_{t+1})))^2$



4. Multi-Objective Optimization

Reward for molecule m and target property p :

$$r_{p,m} = \frac{p_{\text{thresh}} - p_m}{p_{\text{thresh}} - p_{\text{extremum}}}$$

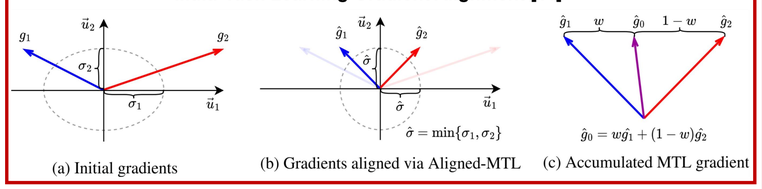
p_{thresh}: user-defined threshold for p $\quad p_m$: p score for m $\quad p_{\text{extremum}}$: extremum of p

Total reward for m with K properties: $R_m = \sum_{k=1}^{K} r_{p_k,m}$

Sweet spot: R_m is doubled if **all properties** are satisfied

Target Property	Target Value	Extremum
Calculated Water-Octanol Partition Coefficient (clogP)	<3 (Minimize)	-3
Synthetic Accessibility Score (SAS)	<3 (Minimize)	1
LPA1 Target Binding Affinity Score (BAS)	<-6 (Minimize)	-5.13 (Lead Avg)

Multi-Task Learning Gradient Alignment [11]



(a) Initial gradients (b) Gradients aligned via Aligned-MTL (c) Accumulated MTL gradient

Results

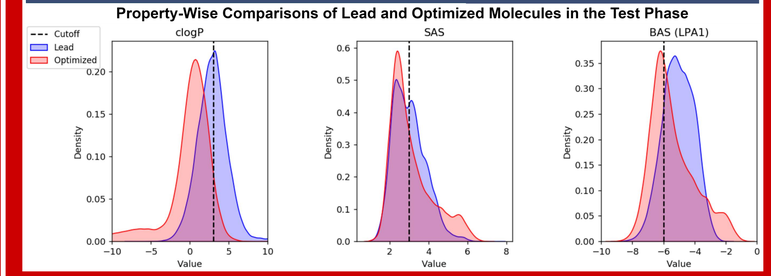
Pre-training Results (~4M Molecules, 80:20 Split, 3 Epochs)

	Training Accuracy (~3.25M Molecules)	Testing Accuracy (~814K Molecules)	Computing Time
MLM	83%	85%	~18 hours
CL	90%	91%	~42 hours

Fine-Tune (115 Epochs, COCONUT + DrugBank Databases: 17K Molecules) & Test Results (COCONUT Database: 118K Molecules)

	Fine-tune Leads: COCONUT + DrugBank (17K mols)	Fine-tune Optimized Molecules	Test Leads: COCONUT (118K mols)	Test Optimized Molecules
Valid	-	91%	-	88%
Novel	-	98%	-	98%
Unique	-	94%	-	87%
1. clogP < 3	54%	85%	53%	82%
2. SAS < 3	55%	65%	50%	51%
3. BAS < -6	21%	43%	20%	38%
Sweet Spot	0%	33%	0%	26%
Computing Time	-	~ 13 Days	-	~ 3 Hours

Property-Wise Comparisons of Lead and Optimized Molecules in the Test Phase



Conclusion and Future Work

The preliminary results are highly encouraging and display the robustness and scalability of our framework in molecular optimization. Ablation studies and benchmarking against other methods will be conducted to conclude this work.